CausalVerify v12 Benchmarks

Exp A : Real Papers — Recognition

Exp B: Synthetic DGPs — Execution

LLMs

Opus | Sonnet | GPT-4o | o3 | Gemini 2.5 | Kimi-128k | GPT-5

Two Matched Task Interfaces

A. Text Pipeline (Exp A)

L1 output → **L3** method → **L4** direction

B. Execution Pipeline (Exp B)

L1 output → **L2a** R code → **L2b** runs → **L2b+** β match

Evaluation, Comparison & Outputs

Evaluation Dimensions

L1	Output exists
L2a	R code present (Exp B)
L2b	Code executes (Exp B)
L2b+	Coefficient match (Exp B)
L3/L4	Text agreement (Exp A)

Core Finding

Text agreement is higher than executable correctness

Exp A	Exp A	Exp B
79%	**81%**	**51%**
L3	**L4**	**L2b+**
Method text	Direction text	Execution

Same seven models; different evidence layers

Outputs & Artifacts

- Reports
- Result figures
- Error analysis
- Benchmarks & code
- Reproducibility package